

RealBackhaulNet: A Real-Data, Constraint-Guarded Neural Policy for Truck-IoT-Assisted Backhaul Decisions

COMPFEST AIC 2026 Haulio Team
AI Innovation Challenge, Indonesia
<https://github.com/ElFariss/compfest-aic-2026-ds>

Abstract—Rule engines and shortest-path solvers are necessary but are not themselves a learned AI product. We present RealBackhaulNet, a from-scratch graph-pointer policy with commercial-vehicle temporal, heavy-truck, deadhead, maintenance, and cost-proxy heads. Seven public sources contribute observed records; rows from unrelated providers are never joined into fictitious trips. A deterministic verifier retains authority over capacity, precedence, compatibility, time windows, road availability, telemetry quality, and unsafe domain shift. The frozen model obtains Amazon next-stop top-1 0.693, Singapore fuel MAE 0.434 in canonical space, and Scania APS average precision 0.747 on deterministic, deadline-bounded held-out subsets. It is exported as checksum-bound TorchScript behind a loopback-only API with reason-coded abstention and mandatory dispatcher approval. Evidence is component-level: no public source supports a causal claim of Haulio operational uplift.

Index Terms—vehicle routing, graph attention, truck telematics, IoT, behavior cloning, domain shift, safe abstention

I. Problem and Research Gap

The challenge requires proactive empty-return risk, ETA, cargo-vehicle matching, multi-hop routing, anomaly detection, and price/margin support using truck IoT. A modified A* or VRP heuristic can solve deterministic search once state, constraints, and cost are known, but it does not learn tacit action preferences or temporal outcomes.

Neural constructive policies learn VRP action sequences [1], [2]; DPDP combines learned edges with exact search [3]. CAADRL adds cluster-aware pickup/delivery attention [4], JAMPR attends jointly to vehicles and customers [5], learned delegation scales local search [6], and PIP studies complex feasibility masks [7]. Their core routing experiments use sampled, generated, or benchmark instances rather than aligned heavy-truck IoT and commercial outcomes. SVRPBench additionally shows substantial degradation under distribution shift [8].

No public source contains the joint tuple of live Haulio telemetry, measured load, current jobs, behavior propensity, dispatcher action, executed route, and realized empty kilometres/margin. Consequently, calling the present system offline RL would be false. We use supervised behavior cloning of real next actions, source-specific auxiliary learning, and explicit abstention.

II. Real Data and Leakage Controls

Table I lists every source entering the checkpoint. Raw files are not redistributed; provider URLs, rights, byte counts, SHA-256 values, processed hashes, sample IDs, and group hashes are recorded in the repository.

Amazon build routes are route-disjoint between train and validation; official evaluation routes are final-only. Candidate neighbourhoods are selected by published-coordinate proximity before checking whether they contain the label. Route scores are conditional on this nearest-32 set containing the published next action and therefore do not measure full-set candidate recall. LaDe splits couriers and includes only tasks accepted by the decision time. Singapore splits vehicles A–F/G–H/I–J. The encoder sees only the first half of a trip and excludes the instantaneous-fuel stream integrated into the fuel target. LaDe ETA intervals are capped at 72 hours. DT-CARGO splits vehicles, VIUS splits registration states, Scania preserves the official test set, and TLC uses days 1–20/21–25/26–31. Final files are first opened after validation checkpoint selection and freezing.

III. Architecture

Candidate nodes $X \in \mathbb{R}^{n \times 16}$ and context $c \in \mathbb{R}^{16}$ map to a 160-dimensional state. Four eight-head graph blocks exchange information among active candidates. A context query q and candidate key k_i give

$$\ell_i = q^T k_i / \sqrt{160} + f_{\text{bias}}([x_i; c]), \quad \ell_i = -10^4 \text{ if } m_i = 0. \quad (1)$$

The pointer is rolled out autoregressively with a stop-after-action auxiliary and ETA-quantile heads; runtime never lets the auxiliary strand a submitted job. A 64×8 Singapore prefix passes through three pre-norm Transformer layers. DT-CARGO, VIUS, APS, and TLC use separate MLP adapters. Cross-family coupling occurs through the shared fleet-state trunk; Amazon and LaDe also share the route branch. Unrelated observations never co-occur in a row.

Route loss combines next-action cross entropy, ETA pinball loss, and STOP cross entropy. Temporal/tabular regression uses masked smooth- L_1 ; idle, home/long-haul, and APS use binary cross entropy; TLC uses ordered

TABLE I
Observed sources used by the frozen checkpoint.

Source	Input unit	Target	Train/val./final	Boundary
Amazon 2021 [9]	route, parcels, windows, cube capacity	actual next driver stop	60,000 / 10,000 / 10,000	U.S. parcel vans; no truck IoT or mass
LaDe [10]	courier tasks, event time/location	actual next task; capped interval	60,000 / 10,000 / 10,000	courier analogue; no truck capacity
Singapore CV [11]	first-half commercial-vehicle OBD prefix	full-trip fuel/duration; future idle	1,831 / 126 / 465	ten vehicles; load label unavailable
DT-CARGO [12]	N3 track, speed, HDOP, mass ratings	duration, signal loss, home/long haul	71,120 / 14,370 / 16,336	cargo and coordinates withheld
VIUS 2021 [13]	truck-use fields plus missing bits	annual mile fractions	17,340 / 4,073 / 3,335	annual prior, not live empty return
Scania APS [14]	170 anonymous fields plus missing bits	APS-related failure	51,085 / 8,915 / 16,000	APS-specific; no vehicle/time ID
NYC TLC [15]	distance, duration, zones, time	metered-fare quantiles	100,000 / 20,000 / 20,000	passenger cost proxy only

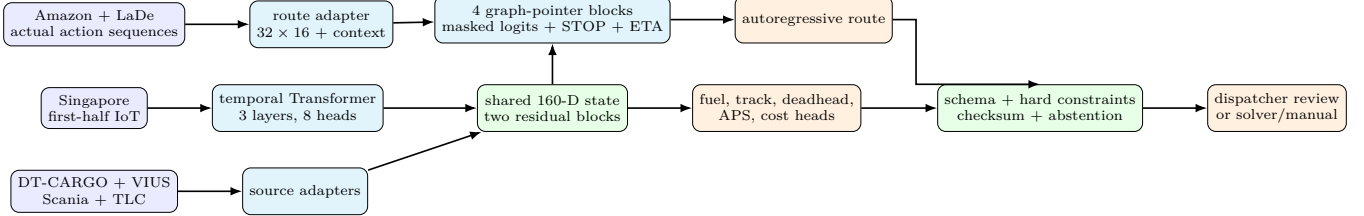


Fig. 1. Real-source training and static deployment architecture. Cross-domain transfer occurs through parameters, not fabricated record joins.

TABLE II
One-time capped final held-out results. Canonical regression MAE remains in documented transformed source space.

Source	Metric	Result
Amazon	Next-stop top-1	0.6934
LaDe	Next-task top-1	0.4364
Singapore	Fuel MAE (canonical)	0.4342
Singapore	Future-idle AP	0.4919
DT-CARGO	Signal-loss MAE (canonical)	0.3391
VIUS	Deadhead MAE (canonical)	0.1728
Scania APS	Failure AP	0.7474
NYC TLC	Fare MAE (canonical)	0.0606

0.1/0.5/0.9 pinball loss. The network has 3,407,060 trainable parameters and is initialized randomly without external weights.

IV. Training and Held-Out Results

Training and frozen evaluation ran on one NVIDIA A100-SXM4-80GB for 40 min 33 s, completing 1,100 source-scheduled minibatches of 192. AdamW, bfloat16 autocast, gradient clipping, cosine decay, a fixed seed, and validation-only checkpoint selection were used. Checkpoint selection uses at most 24 validation batches per source. Final partitions were opened once after freezing; final evaluation is likewise capped at 24 deterministically ordered batches per source (up to 4,608 rows), with exact per-source counts recorded in the metrics ledger.

The nearest-candidate comparator uses exactly the pointer candidate set. The masked-index check is an implementation sanity check, not empirical validation of capacity or legal constraints absent from public route rows. The reloaded TorchScript output differs from eager inference by at most 0.000e+00. Artifact SHA-256 begins 11cb3412afc970a6. These are component metrics, not business KPIs or causal uplift estimates.

V. Runtime Safety and Compliance

The route API requires actual kg/cm³ capacity and load, paired shipment deltas, freshness, fuel/GNSS/operator thresholds, and a complete directed road matrix. Before every action it masks capacity, precedence, vehicle/cargo, window, road, and duration violations. Unverified hazmat/reefer work is conservatively masked. Inconsistent intervals, impossible edges, unsafe domain shift, a stranded pickup, non-finite output, corrupt weights, or unsafe STOP returns ABSTAIN. The fallback is Haulio’s deterministic solver or manual dispatch.

The Compose service binds to loopback, runs unprivileged and read-only, verifies the weight hash at startup, and exposes core inference only. It has no optimizer, backward pass, auto-tuning, bulk-test endpoint, feedback store, checkpoint reload, or dispatch API. Parameters remain static during the competition demonstration.

VI. Limitations and Conclusion

Behavior cloning reproduces observed public decisions, not a counterfactual optimum. Amazon/LaDe are par-

cel domains; Singapore has ten vehicles; DT-CARGO removes cargo/coordinates; VIUS is annual; APS fields are anonymous; TLC is passenger transport. Driver law, hazmat, reefer, axle restrictions, weather, port state, and MQTT identity remain upstream facts. RealBackhaulNet is thus an actual trained AI artifact with honest real-data provenance and a deterministic safety envelope, not a field-validated autonomous dispatcher.

References

- [1] M. Nazari et al., “Reinforcement learning for solving the vehicle routing problem,” NeurIPS, 2018.
- [2] W. Kool et al., “Attention, learn to solve routing problems!” ICLR, 2019.
- [3] W. Kool et al., “Deep policy dynamic programming for vehicle routing problems,” arXiv:2102.11756, 2021.
- [4] W. Wang et al., “Cluster-aware attention-based deep reinforcement learning for pickup and delivery problems,” arXiv:2603.10053, 2026.
- [5] J. Falkner and L. Schmidt-Thieme, “Learning to solve vehicle routing problems with time windows through joint attention,” arXiv:2006.09100, 2020.
- [6] S. Li et al., “Learning to delegate for large-scale vehicle routing,” NeurIPS, 2021.
- [7] J. Bi et al., “Learning to handle complex constraints for vehicle routing problems,” NeurIPS, 2024.
- [8] A. Heakl et al., “SVRPBench: A realistic benchmark for stochastic vehicle routing problem,” NeurIPS Datasets and Benchmarks, 2025.
- [9] D. Merchán et al., “2021 Amazon last mile routing research challenge: Data set,” Transportation Science, 2022.
- [10] L. Wu et al., “LaDe: The first comprehensive last-mile delivery dataset from industry,” arXiv:2306.10675, 2023.
- [11] L. Cheah, “Singapore commercial vehicle GPS, OBD, payload, fuel use data,” Figshare, 2019, doi: 10.6084/m9.figshare.9741035.v2.
- [12] Technical University of Munich FTM, “Dataset of trucks’ anonymized recorded driving and operation,” Zenodo, 2025, doi: 10.5281/zenodo.16411298.
- [13] U.S. Census Bureau, “2021 Vehicle Inventory and Use Survey Public Use File,” 2023. [Online]. Available: <https://www.census.gov/data/datasets/2021/econ/vius/2021-vius-puf.html>
- [14] Scania CV AB, “APS Failure at Scania Trucks,” UCI Machine Learning Repository, 2017, doi: 10.24432/C51S51.
- [15] New York City Taxi and Limousine Commission, “Yellow Taxi Trip Records, January 2024,” 2024. [Online]. Available: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>